

Fazle Rabbi Zaki

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US Citizen

Education

University of Michigan

B.S.E. in Aerospace Engineering

Ann Arbor, MI

Expected May 2026

- University Honors (103rd Annual Honors Convocation); Dean's Honor List: Fall 2025
- Coursework: Attitude Determination & Control Systems (grad), Engineering for Space Environment (grad), Automatic Control, Space System Design, Spacecraft Dynamics, Propulsion, Aerodynamics

Technical Skills

- GNC/ADCS:** Rigid-body dynamics, quaternion kinematics, B-dot detumble, PID/LQR control, state estimation (TRIAD, EKF), momentum management, hardware-in-the-loop testing
- Software:** Python (NumPy, SciPy, pandas), MATLAB/Simulink, C++, Git, Linux
- CAD/Analysis:** SolidWorks, CATIA, STAR-CCM+ (CFD)
- Embedded/Lab:** Arduino, Raspberry Pi, UART/I2C/SPI, oscilloscope, DAQ systems, multimeter, soldering

Engineering Projects

CubeSat ADCS Imaging Testbed

Ann Arbor, MI

ADCS & Systems Lead | University of Michigan (Graduate-Level Course)

Aug 2025 – Dec 2025

- Designed and integrated 1U-class CubeSat prototype (0.95 kg) with reaction wheel, torque rods, IMU, and Raspberry Pi camera modules for autonomous detumble-to-imaging operations
- Developed MATLAB EKF (angle + rate states) and TRIAD-based attitude estimation; evaluated sensor/actuator resolution limits and proposed 12-bit motor control to meet 0.001 rad/s rate step requirement
- Led systems engineering for 15-member team: derived Level-1 requirements, maintained mass/power/cost budgets, and delivered \$3,000 total system cost vs \$6,000 cap
- Designed Helmholtz coil HIL testbed producing $\geq 3\times$ Earth field strength (168 μT) with $\leq 10\%$ reproducibility; demonstrated spin stoppage in 50 s during functional test

Icarus Heliophysics Mission

Ann Arbor, MI

Systems Engineering & ADCS | Safety, Policy & Risk Lead

Jan 2026 – Apr 2026

- Conducted safety, policy, and risk analysis for a dual-spacecraft heliophysics mission through NASA Phase B PDR; mapped 12 rights-holders and identified 3 mission-critical risks including SEP transit exposure and launch window constraints
- Contributed to attitude determination sensor trade study leveraging graduate-level ADCS coursework; evaluated sun sensors, magnetometers, star trackers, and IMUs against accuracy, mass, power, and cost metrics for a zero-momentum architecture
- Supported systems engineering including requirements flowdown from mission-level objectives to subsystem specifications, mass/power budgeting, and interface definition across 10 subsystems
- Ensured regulatory compliance with NPR 8715.6 (debris mitigation), ITAR/DDTC (export control), and FCC/ITU (spectrum allocation) for deep-space X/Ka-band communications

Horizontal-Axis Wind Turbine Generator

Ann Arbor, MI

Design Engineer | University of Michigan

Jan 2024 – Apr 2024

- Designed and built self-yawing 1.54 m HAWT (three 0.80 m blades) on \$110 budget; generated 6.45 V peak output during rooftop tests
- Verified structural integrity through five safety inspections, including 300 N lateral load test and high-wind disassembly, enabling rooftop deployment approval
- Instrumented and analyzed performance (TSR up to 7.2); diagnosed drivetrain losses and recommended 1:3 gearing to increase generator shaft speed

Titan Air Puck

Ann Arbor, MI

Design Engineer | University of Michigan

Jan 2023 – Apr 2023

- Modeled hexagonal lift platform in CATIA and validated with STAR-CCM+ CFD to optimize lift-to-power ratio, simulating 1.41 N lift at 1.51 W input
- Built and tested prototype at 269 g system mass (< 800 g constraint); demonstrated 6.68 s minimum crossing time and lifted 80 g payload in competition
- Executed design trade studies on plenum/skirt geometry and materials to improve sealing and lift stability for Titan-environment concept

Activities

Michigan Aeronautical Science Association (MASA)

Ann Arbor, MI

Avionics Subteam Member

Aug 2025 – Present

- Integrating avionics subsystems for hybrid rocket targeting 30,000 ft apogee; contributing to cold-flow tests and HITL infrastructure for sensor/pyro telemetry verification
- Configuring ground support equipment and executing E-Box sensor checkout for propulsion testing at MCity facility